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On the Kauffman polynomial of an adequate link

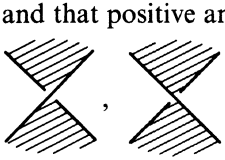
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1. Introduction

In the recent article [6], the concept of an *adequate link diagram* was introduced, and investigated in the context of the Jones polynomial. Here we use the Kauffman polynomial invariant $F(a, z)$ of semi-oriented link types [3] to show that adequate diagrams share some of the desirable properties enjoyed by the narrower class of reduced alternating diagrams, in particular minimality of crossing-number and invariance of writhe. The weaker property *semi-adequacy* is also investigated, and it is shown that any reduced semi-adequate diagram represents a non-trivial link. Moreover, we show that any connected semi-adequate diagram represents a non-split link; this generalizes Theorem 1 a) of [7] (proved earlier by different methods in [1] and [2]). These results are all straightforward consequences of Theorem 1 below, which states a relationship between the regular isotopy form $\Lambda(a, z)$ of Kauffman's polynomial, and the Tutte polynomial. This relationship was hinted at in [10], but here arguments of a more involved nature are used. It is interesting that the Tutte polynomial also manifests itself in a different way in $F(a, z)$, namely in the Jones polynomial, of which $F(a, z)$ is a generalization [5, 9].

2. Graph-theoretical background

Although it is by no means mandatory, we shall usually work with the connected planar graph G, with signed edges, associated with a choice of chequerboard colouring of the regions of a connected link diagram D, rather than with the diagram itself. The convention taken will be that the vertices of G correspond to the black regions of D, and that positive and negative edges of G correspond to crossings in D of types  respectively.

We introduce some notation. Given a graph G associated with a connected link diagram, G_+ is the subgraph of G consisting of all the vertices of G and the positive edges of G; the "quotient" graph $\overline{G_+}$ is obtained from G_+ by identifying those pairs of vertices which are ends of a path consisting of *negative* edges of G. The graphs G_-, $\overline{G_-}$ are defined likewise. We note that the graphs $\overline{G_+}$, $\overline{G_-}$ are connected. Note also that the graphs G_+, $\overline{G_-}$ have the following "dual" relationship to one another: G_+ is obtained from G by deleting all negative edges of G, whereas $\overline{G_-}$ is obtained by contracting all positive edges of G.

For any graph G, the number of components of G is denoted $p_0(G)$; the *cyclomatic number* $p_1(G)$ of G is the least number of edges of G whose deletion destroys all the circuits of G. If we regard G as being a 1-dimensional complex, then $p_0(G)$, $p_1(G)$ are the ranks of the homology groups $H_0(G)$, $H_1(G)$ respectively. An edge of a graph G will be said to be *essential* in G if it is neither an isthmus of G nor a loop of G. Essential edges of G correspond to non-nugatory crossings of the associated link diagram, i.e. crossings incident to four distinct regions of the diagram. If e is an edge of G, then G'_e, G''_e, G^σ_e will denote the graphs obtained by respectively deleting e, contracting e, switching the sign of e. If the case of G''_e it is assumed that e is not a loop.

We shall use the graph-theoretical terminology of sections 2, 3 of [9], but we shall need to adjust two definitions, so as to cater for slightly more general situations. The first such case is that of a *block* of a graph, which was defined in [9] for loopless graphs. We shall extend this definition to graphs containing loops by considering that each loop of a graph G together with its incident vertex shall constitute a block of G. The other blocks of G are precisely the blocks, according to the more restrictive definition, of the loopless graph obtained by deleting all loops of G. If G represents a link diagram D, then the blocks of G correspond bijectively to the connected summands of D.

Secondly, we shall need to consider the Tutte polynomial $\chi_G(x, y)$ of a graph with more than one component. The Tutte polynomial of this more general type of graph is simply defined to be the product of the Tutte polynomials of its components. According to this definition, the adjunction of some isolated vertices to G does not affect χ_G. Also, this extended polynomial satisfies the usual identity $\chi_G(x, y) = \chi_{G'_e}(x, y) + \chi_{G''_e}(x, y)$, it being assumed that the edge e is not an isthmus or loop of G.

A further remark concerning the Tutte polynomial is that it is well-known (see for instance [11]) that for any graph G, $\chi_G = \chi_{G_1} \cdots \chi_{G_k}$, where $G_1, \ldots, G_k$ are the blocks of G.

3. The relationship between $\Lambda(a, z)$ and the Tutte polynomial

We recall the basic properties of the polynomial $\Lambda(a, z)$. Λ is an invariant of regular isotopy types of unoriented diagrams; if D is a diagram with writhe w of an *oriented* link L, then $\Lambda_D(a, z) = a^w F_L(a, z)$, where F_L is the Kauffman polynomial of L. As usual, let us assume that a chequerboard colouring of the regions of the diagram D has been chosen. The polynomial $\Lambda_D(a, z)$ is characterized by the following list of axioms:

(i) Λ_D is a regular isotopy invariant of diagrams of L;

(ii) $\Lambda_D = 1$ if D is a simple closed curve;

(iii) (the *kink* formulae) $\Lambda_{D_1} = a\Lambda_D$ and $\Lambda_{D_2} = a^{-1}\Lambda_D$, where D_1, D_2 are obtained from D by adjoining a positive kink and a negative kink respectively, as in Fig. 1a;

(iv) (the *switching* formula) $\Lambda_{D_+} + \Lambda_{D_-} = z(\Lambda_{D_0} + \Lambda_{D_\infty})$, where D_+, D_-, D_0, D_∞ are diagrams which are identical except in the vicinity of a single crossing, where they differ as shown in Fig. 1b.

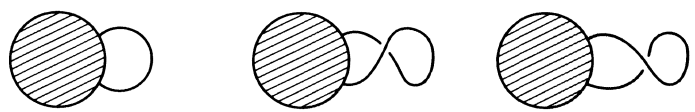


Fig. 1a

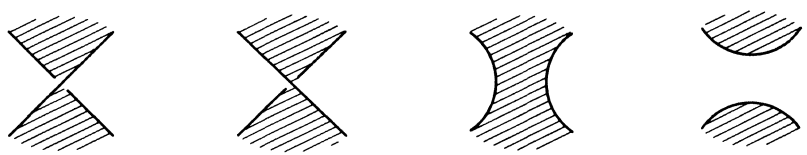


Fig. 1b

These definitions of D_+, D_-, D_0, D_∞ depend on a choice of chequerboard colouring of D, but the symmetry of the switching formula ensures that Λ_D does not depend on this choice.

One of the difficulties in understanding the Λ-polynomial of a diagram $D = D_+$ arises from the presence of Λ_{D_-} in the switching formula. M.E. Kidwell, in [4], showed how, in certain circumstances, one could ignore the contribution of Λ_{D_-}, so as to obtain useful information on terms of uppermost degree in the one-variable polynomial $Q_L(x) = F_L(1, x)$. Using a development of this idea, it was shown in [10] that if we write $\Lambda_D(a, z) = \Sigma u_{rs} a^r z^s$, and if the diagram D has n crossings, then u_{rs} can only be non-zero if $r + s \leqq n$ and $-r + s \leqq n$. It will be shown below (Propositions 4, 5) that if suitable conditions prevail, then one or other of these inequalities is strict; it is this that provides the basis for the proof of Theorem 1. In view of the constraints $r + s \leqq n$, $-r + s \leqq n$ on the "coordinates" r and s, we shall call a coefficient u_{rs} *outermost* if $r + s = n$ or $-r + s = n$. It will be convenient to encapsulate these outermost coefficients in two separate one-variable polynomials $\phi_D^+(t)$, $\phi_D^-(t)$ defined as follows:

$$\phi_D^+(t) = \sum_i u_{i,n-i} t^i, \quad \phi_D^-(t) = \sum_i u_{-i,n-i} t^i.$$

From the preceding remarks, there are no negative powers of t occurring in $\phi_D^+(t)$ or $\phi_D^-(t)$. Later in the article, when considering the graph G associated with D we shall sometimes write $\phi_G^\pm$ instead of $\phi_D^\pm$.

We can now state the fundamental result of this paper; section 5 will be devoted to its proof.

Theorem 1. *Let D be a connected link diagram with at least one crossing, with associated graph G. Then*

$$\phi_D^+(t) = \chi_{G_+}(0,t)\,\chi_{\overline{G_-}}(t,0), \quad \text{and} \quad \phi_D^-(t) = \chi_{G_-}(0,t)\,\chi_{\overline{G_+}}(t,0).$$

Using the basic properties of the Tutte polynomial explained in § 3 of [9], we can immediately obtain specific information on the nature of $\phi_D^+(t)$, $\phi_D^-(t)$. Corollary 1.1 is stated for $\phi_D^+(t)$, but obviously reversal of signs will yield corresponding results for $\phi_D^-(t)$.

Corollary 1.1. *Let D be a connected link diagram with at least one crossing.*

(i) *All coefficients of $\phi_D^+(t)$ are non-negative.*

(ii) *$\phi_D^+(t) \neq 0$ if and only if G_+ has no isthmus and $\overline{G_-}$ has no loop.*

(iii) *If $\phi_D^+(t) \neq 0$, the term of lowest degree in $\phi_D^+(t)$ is κt^b, where $\kappa \geqq 1$ and b is the total number of blocks which are not isolated vertices in the graphs G_+, $\overline{G_-}$.*

(iv) *If $\phi_D^+(t) \neq 0$, the term of highest degree in $\phi_D^+(t)$ is $t^{p_1(G_+)+a-1}$, where a is the number of vertices of $\overline{G_-}$.*

(v) *The degree of $\phi_D^+(t)$ does not exceed the number of crossings of D, and is strictly less than this number of crossings if D has a non-nugatory crossing.*

Proof. (i) From the definition of the Tutte polynomial given in [9], each coefficient in a Tutte polynomial is non-negative.

(ii) As explained in § 3 of [9], if a graph G contains an isthmus, then $\chi_G(x,y)$ is divisible by x, and if G contains a loop, then $\chi_G(x,y)$ is divisible by y. The converse of this assertion follows from the proof of Proposition 2 (ii) of [9].

(iii) If $\phi_D^+(t) \neq 0$, from part (ii) each block B of G_+, which is not an isolated vertex, is either a single loop together with its incident vertex, in which case $\chi_B(x,y) = y$,or else B is a loopless non-separable graph with at least one cycle, in which case the coefficient of y in $\chi_B(x,y)$ is strictly positive by Proposition 3(i) of [9]. Similarly, if B is a block in $\overline{G_-}$, the coefficient of x in $\chi_B(x,y)$ is strictly positive. On the other hand, for any connected graph B with at least one edge, the coefficient of x^0y^0 in $\chi_B(x,y)$ is zero, as the lowest ranking edge of B is active in any spanning tree of B.

(iv), (v) These follow from Proposition 2 of [9]; note that the coefficient of the highest power of t is always 1. □

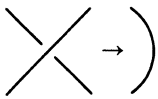
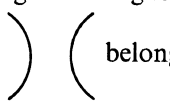
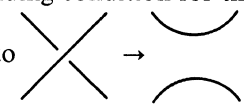
The reader can check without difficulty that the statements of Theorem 1 and Corollary 1.1 are invariant under reversal of the colour-scheme of the regions of the diagram D. The statement that $\overline{G_-}$ has no loop is equivalent to the statement that there is no cycle of G containing just one negative edge.

Admittedly, Theorem 1 merely "scratches the surface" of the polynomial Λ. It would be interesting to see if a refinement of the techniques presented here could be used to penetrate further into this polynomial.

J. Przytycki has kindly pointed out that the formula of Theorem 1 uses, up to a change of variable, the *chromatic* polynomials (see [11]) of the graphs $\overline{G_\pm}$, and the duals of the graphs $G_\pm$. It then follows from known properties of chromatic polynomials that the computation of Λ, even for alternating links, is NP-hard.

4. Applications to adequate and semi-adequate diagrams

We begin by defining a one-sided version of "adequacy".

Definition. A link diagram D is $+$*adequate* if (i) D has at least one crossing, and (ii) in the Kauffman state $s_+ D$ formed by nullifying each crossing according to the rule  , for each crossing the two segments  belong to different components of $s_+ D$. If D satisfies the corresponding condition for the state $s_- D$ obtained by nullifying each crossing according to  , then D is $-$*adequate*.

Clearly, a diagram D is $+$adequate if and only if its obverse (obtained by switching all crossings) is $-$adequate.

Definition. A diagram D is *adequate* if it is both $+$adequate and $-$adequate, and is *semi-adequate* if it is either $+$adequate or $-$adequate. D is $+$*inadequate* ($-$*inadequate*) if it is not $+$adequate (not $-$adequate), and D is *inadequate* if it is neither $+$adequate nor $-$adequate.

The next Proposition translates the notion of adequacy into graph-theoretical language. As in [6], given a Kauffman state sD of a diagram D, $|sD|$ denotes the number of components of sD.

Proposition 2. *Let G be the graph of a chequerboard colouring of a connected diagram D.*

(i) $|s_+ D| = p_0(G_+) + p_1(G_+)$ *and* $|s_- D| = p_0(G_-) + p_1(G_-)$.

(ii) *D is $+$adequate if and only if G_+ has no isthmus and $\overline{G_-}$ has no loop, and D is $-$adequate if and only if G_- has no isthmus and $\overline{G_+}$ has no loop.*

Proof. Part (i) follows easily by induction on the number of crossings of D (edges of G), the inductive step being achieved by considering what happens when a non-nugatory crossing of D is nullified in the two possible ways. For part (ii), recall from [6] that D is $+$adequate if and only if $|s_+ D^\sigma| < |s_+ D|$ for each diagram D^σ obtained from D by switching a single crossing. Now, changing the sign of an edge of G will either increase by one or decrease by one precisely one of $p_0(G_+)$, $p_1(G_+)$. The condition on the graph of a $+$adequate diagram, given in part (ii) of the statement, is just what is needed to ensure that changing the sign of a single edge of G does not increase the number of components of G_+ nor increase its cyclomatic number. The $-$adequate case is dealt with similarly. □

The applications of Theorem 1 to adequate and semi-adequate diagrams can now be stated.

Theorem 3. *Let D be a link diagram with at least one crossing.*

(i) $\phi_D^+(t) \neq 0$ *if and only if D is $+$adequate, and $\phi_D^+(t) \neq 0$ if and only if D is $-$adequate.*

(ii) *If D has a non-nugatory crossing and is semi-adequate, then the degree in z of $\Lambda_D(a,z)$ is strictly positive; hence D represents a non-trivial link.*

Proof. First, let us consider the case where the diagram D is connected. In this case, part (i) follows immediately from Corollary 1.1 (ii) and Proposition 2; part (ii) then follows from Corollary 1.1(v). If, on the other hand, D is not connected, the result follows from the special case in conjunction with the formula $\Lambda_D = ((a^{-1}+a)z^{-1}-1)^{c-1}\Lambda_{D_1}\ldots\Lambda_{D_c}$ for the Λ-polynomial of D in terms of the Λ-polynomials of its components $D_1,\ldots,D_c$. □

Part (ii) of Theorem 3 constitutes an exceptionally simple, yet effective test for non-triviality of a link. A measure of its effectiveness is that all diagrams of minimal crossing-number of knots of up to 11 crossings are semi-adequate, as are all but a handful of diagrams of minimal crossing-number of 12-crossing knots. An example of a 12-crossing knot with no semi-adequate 12-crossing diagram is shown in Fig. 2. This knot is amphicheiral; I do not know whether it admits a semi-adequate diagram with more than 12 crossings.

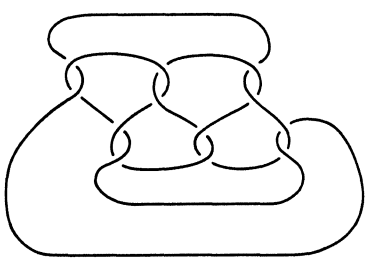


Fig. 2. A 12-crossing knot with no 12-crossing semi-adequate diagram

Corollary 3.1. *A semi-adequate diagram is not regularly isotopic to a diagram with fewer crossings.*

Proof. By Theorem 3(i), Λ_D has a non-zero outermost coefficient; in view of the constraints $r+s \leqq n$, $-r+s \leqq n$ on the non-zero coefficients u_{rs} of Λ_D, there cannot exist a diagram of fewer crossings with the same Λ-polynomial. □

Corollary 3.2. *A connected semi-adequate diagram cannot represent a split link.*

Proof. Suppose L is a split link with split components $L_1,\ldots,L_c$, presented as a "realization" of a connected, n-crossing diagram$D$ in a suitably small region of a 2-sphere S^2, i.e. L coincides with the edges of D except for small semicircular open arcs in S^3-S^2 near the crossing-points of D. Let $D_1,\ldots,D_c$ be the sub-diagrams of D representing the sublinks $L_1,\ldots,L_c$ respectively. Then L is isotopic to a similarly presented link L', the diagram of L' being a disjoint union of diagrams isomorphic to $D_1,\ldots,D_c$. This diagram D' of L' has at most $n-2$ crossings, since it has only the self-crossings of $D_1,\ldots,D_c$. Also, since each component of L_i has zero linking number with each component of $L_j (i \neq j)$, D' has the same writhe as D, and so D' is regularly isotopic to D and Corollary 3.1 applies. □

Corollary 3.3 *Let D_1, D_2 be n-crossing diagrams of a link L, with writhes w_1, w_2 respectively. If D_1 is +adequate, then $w_1 \geqq w_2$, and if D_1 is −adequate, then $w_1 \leqq w_2$. If D_1 is adequate, then D_2 is also adequate and $w_1 = w_2$.*

Proof. If D_1 is +adequate, then by Theorem 3(i) $\phi^+_{D_1}(t) \neq 0$; hence an n-crossing diagram with writhe greater than that of D_1 would have a Λ-polynomial equal to $a^t \Lambda_{D_1}(a, z)$ for some $t \geqq 1$, violating the constraint $r + s \leqq n$ on the non-zero coefficients u_{rs} of its Λ-polynomial. The conclusions concerning −adequate and adequate diagrams are dealt with similarly. □

The proof of the next Corollary is very similar to that of Corollary 3.1, and is therefore omitted.

Corollary 3.4. *If a link L has an adequate diagram with n crossings, then L cannot be projected with fewer than n crossings.* □

Corollary 3.5. *For each natural number n there exists a prime diagram which is of minimal crossing-number for the link it represents, yet which contains a bridge of length greater than n.* (No claim is made regarding the primality of the *links* thus represented.)

Proof. Consider the infinite family of graphs $G_1, G_2, \ldots$, of which the first three members are illustrated in Fig. 3. Within each graph G_i, the subgraph $(G_i)_+$ is to consist of the left-hand family of nested circles. Each G_i corresponds to an adequate diagram by Proposition 2(ii), which is of minimal crossing-number for the link it represents by Corollary 3.4, yet the $2i+1$ emphasized edges in Fig. 3 correspond to crossings belonging to a bridge of length $2i+1$. For good measure, each G_i can be changed into a graph corresponding to an adequate diagram of a knot with a bridge of length $2i+1$, if desired, by judiciously inserting extra vertices in the middle of edges. □

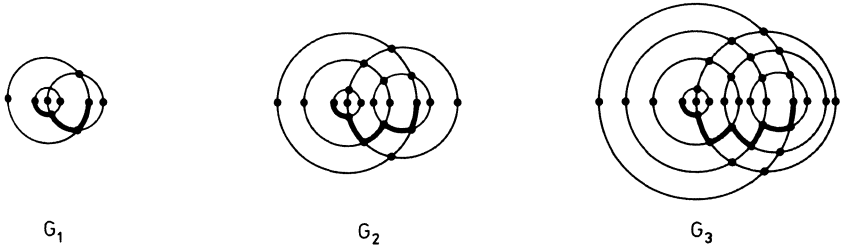


Fig. 3.

Example: the 10-crossing knot listed as 10_{161}, 10_{162} in [8]. This knot is somewhat notorious, as it admits a 10-crossing diagram with writhe 8, and another 10-crossing diagram with writhe 10. These diagrams, illustrated in Fig. 4 together with their associated graphs, were thought to represent distinct knot types, until K.A. Perko spotted the equivalence between them in 1974.

From Theorem 1,

$$\phi^+_{D_1}(t) = 0 \quad \text{and} \quad \phi^-_{D_1}(t) = t^2(t^2 + t^3),$$
$$\phi^+_{D_2}(t) = t^2 \cdot t^2 \text{ and } \quad \phi^-_{D_2}(t) = 0.$$

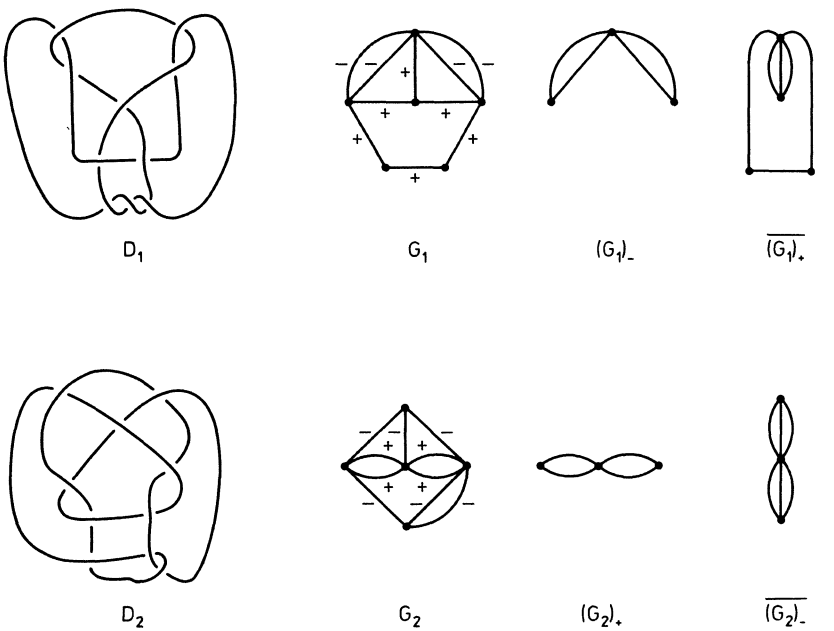


Fig. 4.

The complete polynomial $\Lambda_{D_1}(a, z) = \Sigma u_{rs} a^r z^s$ may conveniently be set out as the following array of integers in the (r, s)-plane; bearings are determined by the statement that the top row of the array corresponds to $(a^{-4} + a^2) z^6$.

	1		0		0		1
1		1					
	−4		1		−1		−6
−4		−3		0		−1	
	3		−3		3		9
3		1		0		2	
			1		−1		−3

5. Proof of Theorem 1

We begin with a rather delicate Proposition, which is the linchpin of the proof of Theorem 1. It will be used in the case of a bridge of length 3 for the proof of Proposition 5, but we need the more general statement, in order to have induction at our disposal.

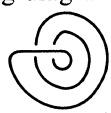
Proposition 4. *Let D be a connected diagram with the following properties:*

(i) *D is reduced, i.e. D contains no nugatory crossing;*

(ii) *D contains a bridge β of length greater than 1;*

(iii) *an alternating diagram can be obtained from D by switching alternate crossings within β.*

Then $\phi_D^+(t) = \phi_D^-(t) = 0$.

Note that there exists a 2-crossing diagram D fulfilling parts (ii), (iii) of the hypothesis, with $\phi_D^+(t) \neq 0$, namely  . Also, Corollary 3.5 shows that part (iii) of the hypothesis cannot be dropped.

Proof of Proposition 4. If D is a reduced diagram satisfying the hypothesis, we define the *complexity* of D to be the ordered pair (n, k), where n is the number of crossings of D and k is the number of crossing switches required to change D to a diagram D' satisfying the hypothesis, but with a bridge β' of maximal length containing β (by "maximal" it is meant that the bridge β' cannot be elongated by crossing switches). Complexities will be ordered lexicographically.

We proceed now by induction on the complexity of D. If $k = 0$, then β is maximal and D is regularly isotopic to a diagram with fewer crossings; therefore $\phi_D^+(t) = \phi_D^-(t) = 0$ from the constraints on r, s for non-zero coefficients u_{rs} in Λ_D. This situation is illustrated in Fig. 3(ii), (iii) of [10]. Also, it is easy to check that the conclusion holds for diagrams with the least possible value of n, namely $n = 2$ (such a diagram would be a 2-crossing diagram of the unlink of two components). Suppose therefore that D is a diagram, satisfying the hypothesis, for which $n > 2$ and $k = 0$, and that the conclusion holds for all diagrams of lower complexity. Then there is a crossing x of D, adjacent to one end of the bridge β, the switching of which will result in a diagram D_- with lower complexity than D. Let D_0, D_∞ be the $(n-1)$-crossing diagrams formed by nullifying the crossing x. The proof will be completed by application of the switching formula for Λ, once we have shown that $\phi_{D_0}^+ = \phi_{D_0}^- = 0$ and $\phi_{D_\infty}^+ = \phi_{D_\infty}^- = 0$. Without any loss of generality, we need only consider D_0. This diagram D_0 is connected, since the crossing x was not nugatory. We shall consider various cases.

If D_0 contains no nugatory crossing, then D_0 fulfils the hypothesis of the Proposition, so $\phi_{D_0}^+ = \phi_{D_0}^- = 0$ by the inductive hypothesis. If a crossing y of D, in the bridge β but not adjacent to x, becomes nugatory in D_0, then there is a simple closed curve C meeting the arcs of D transversely in four points, as shown in Fig. 5a. Then D is regularly isotopic to a diagram with fewer crossings, so in any event $\phi_D^+ = \phi_D^- = 0$.

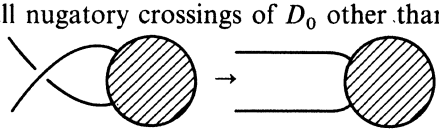
Next, suppose that the crossing z of β adjacent to x becomes nugatory in D_0, and that the other crossings of β do not become nugatory in D_0. We shall see that we can then reduce D_0 by removing nugatory crossings, to a diagram D_0' which either fulfils the hypothesis of the Proposition and has lower complexity than D, or which satisfies $\phi_{D_0'}^+ = \phi_{D_0'}^-$ on account of being regularly isotopic to a diagram with fewer crossings. First, we remove all nugatory crossings of D_0 other than z by untwisting in the obvious way, i.e.  , thus



Fig. 5a.

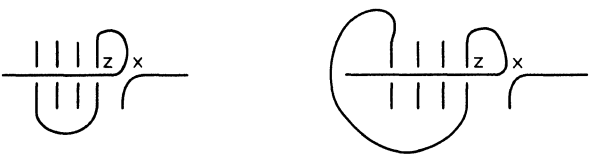


Fig. 5b.

creating a diagram D_0'' say. This untwisting process cannot affect the non-nugatory status of other crossings. If there is an arc of D_0'' joining the underpass of z to an underpass of the bridge β, as in Fig. 5b, then D_0'' is regularly isotopic to a diagram with fewer crossings, and we choose D_0' to be D_0''. Otherwise, part (iii) of the hypothesis guarantees that the diagram obtained from D_0'' by untwisting *z has a bridge with length equal to that of β*; hence this diagram satisfies the hypothesis of the Proposition, and has lower complexity than D. In this event, this diagram, which in effect has been obtained by removing all nugatory crossings from D_0, is deemed to be D_0'. Now we reconstruct D_0 from D_0' by reinstating the nugatory crossings. Each time a nugatory crossing is introduced, the Λ-polynomial is multiplied by a or a^{-1}, and the number of crossings is increased by one; hence the property $\phi^+ = \phi^- = 0$ is preserved at each stage. We conclude that $\phi^+_{D_0} = \phi^-_{D_0} = 0$, and the conclusion therefore holds in this case.

A simplified version of the argument in the last paragraph deals with the remaining case, where no crossing of β becomes nugatory in D_0, but where one or more crossings not on β become nugatory in D_0. Since all eventualities have been catered for, the Proposition is proved. □

Proposition 5. *Let D be a connected, $+$inadequate (respectively $-$inadequate) diagram with $n > 0$ crossings. Then $\phi^+_D = 0$ (respectively $\phi^-_D = 0$).*

Proof. The proof for $-$inadequate diagrams is the same as that for $+$inadequate diagrams with all signs reversed.

Let D be a connected, $+$inadequate diagram with $n > 0$ crossings, and with graph G determined by some chequerboard colouring of the regions of D. We proceed by induction on the number k of edges of G_+ which are not isthmuses or loops of G. If $k = 0$, then G contains a positive isthmus or a negative loop, corresponding to a nugatory crossing of negative writhe in D, as G is $+$inadequate; since removal of this nugatory crossing has the effect of reducing the number of crossings of D by one and multiplying $\Lambda_D(a, z)$ by a, $\phi^+_D = 0$ follows from the constraints on r, s for non-zero coefficients u_{rs} of Λ_D. If $k = 1$, let D_1 be the diagram obtained from D by untwisting all nugatory crossings. D_1 fulfils the

hypothesis of Proposition 4, so $\phi^+_{D_1} = 0$; that $\phi^+_D = 0$ then follows from the argument given near the end of the proof of Proposition 4.

Suppose now that G has $k > 1$ positive edges essential in G, and that the conclusion holds for all connected, +inadequate diagrams with less than k such edges. Let $\mathscr{E}$ denote the set of these k edges.

First let us consider the case where $\mathscr{E}$ has an edge e essential in G_+. From the condition given in Proposition 2(ii), contraction of e does not destroy +inadequacy, so G''_e is +inadequate. We must check that G'_e, G^σ_e are also +inadequate, so that we can get the desired conclusion from the inductive hypothesis in conjunction with the switching formula for Λ. The only possible cause for concern here is that G_+ might have no isthmus, and each cycle containing just one edge of G_- might contain e. However, this cannot happen: since e is essential in G_+, e belongs to a cycle in G_+, and so any cycle in G containing just one negative edge and containing e can be replaced by a cycle containing just one negative edge and not containing e. Therefore the conclusion follows in this case.

If, on the other hand, $\mathscr{E}$ consists entirely of $k > 1$ isthmuses of G_+, it is clear from Proposition 2 (ii) that G'_e, G''_e, G^σ_e are all +inadequate, for any choice of edge e in $\mathscr{E}$. Therefore the inductive hypothesis and the switching formula give the required result in this case also. □

The next Proposition is an immediate consequence of Proposition 2(ii); its proof is therefore omitted.

Proposition 6. *Let G be a +adequate (respectively −adequate) graph, and let e be any edge of G. Then G^σ_e is −inadequate (respectively −inadequate).* □

Proof of Theorem 1. We shall only prove the formula for $\phi^+_D(t)$, as a parallel argument will prove the formula for $\phi^-_D(t)$. We proceed by induction on the number of edges of G.

Suppose first that the connected diagram D is +inadequate. From Proposition 5, $\phi^+_D(t) = 0$, and from Proposition 2(ii) either G_+ has an isthmus, in which case $\chi_{G_+}(0,t) = 0$, or $\overline{G_-}$ has a loop, in which case $\chi_{\overline{G_-}}(t,0) = 0$. Therefore the formula for $\phi^+_D(t)$ given in the Theorem holds in this case.

Next, let us consider the extreme case where D is +adequate, but neither G_+ nor $\overline{G_-}$ contains an essential edge. Then, from Proposition 2(ii), each edge of G_+ is a loop, and each edge of $\overline{G_-}$ is an isthmus. From the manner in which $\overline{G_-}$ was constructed, it follows that $G_- = \overline{G_-}$, and that each edge of G_- is an isthmus of G. Therefore the set of edges of G consists of p positive loops and q negative isthmuses say, and so $\chi_{G_+}(0,t)\,\chi_{G_-}(t,0) = \chi_{G_+}(0,t)\,\chi_{\overline{G_-}}(t,0) = t^{p+q}$. Now this graph G represents a diagram of the unknot with writhe $p+q$; from the kink formula for Λ, $\Lambda_D(a,z) = a^{p+q}$, and from the definition of ϕ, $\phi^+_D(t) = t^{p+q}$. Therefore the formula for $\phi^+_D(t)$ holds in this case.

Finally, let us suppose that D is +adequate, and that G_+ or $\overline{G_-}$ contains an essential edge e, say. We shall deal only with the case where e is essential in G_+, as the case $e \in \overline{G_-}$ is dealt with similarly. From Propositions 5 and 6, $\phi^+_{G^\sigma_e} = 0$; therefore, from the switching formula, $\phi^+_G(t) = \phi^+_{G'_e}(t) + \phi^+_{G''_e}(t)$. It is easily checked straight from the relevant definitions that $(G'_e)_+ = (G_+)'_e$, $(G''_e)_+ = (G_+)''_e$,

and that $\overline{(G''_e)_-}$ is isomorphic to $\overline{G_-}$. Also, since e is essential in G_+, $\overline{(G'_e)_-}$ is isomorphic to $\overline{G_-}$. Then

$$\begin{aligned}\phi^+_G(t) &= \phi^+_{G'_e}(t) = \phi^+_{G''_e}(t)\\ &= \chi_{(G'_e)_+}(0,t)\,\chi_{\overline{(G'_e)_-}}(t,0) + \chi_{(G''_e)_+}(0,t)\,\chi_{\overline{(G''_e)_-}}(t,0)\\ &\quad \text{(by the inductive hypothesis)}\\ &= \chi_{(G_+)'_e}(0,t)\,\chi_{\overline{G_-}}(t,0) + \chi_{(G_+)''_e}(0,t)\,\chi_{\overline{G_-}}(t,0)\\ &= (\chi_{(G_+)'_e}(0,t) + \chi_{(G_+)''_e}(0,t))\,\chi_{\overline{G_-}}(t,0)\\ &= \chi_{G_+}(0,t)\,\chi_{\overline{G_-}}(t,0), \text{ by the recurrence relation for } \chi. \quad \square\end{aligned}$$

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